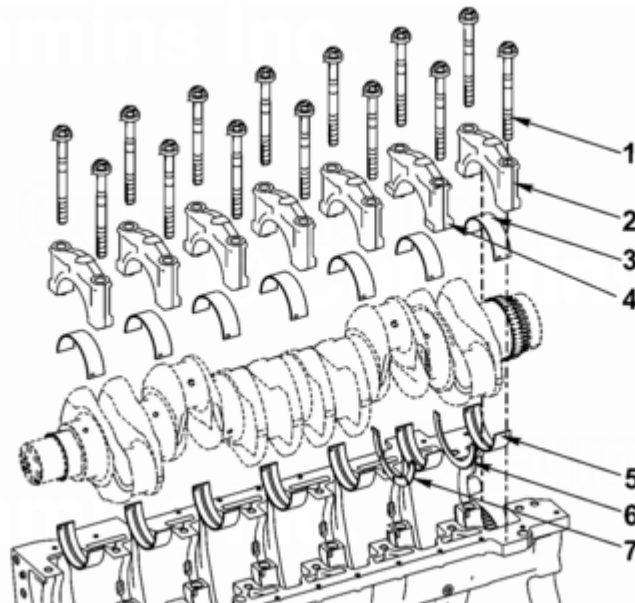


Trainings ▾

Exploded View



Main Bearings Exploded View

1. Capscrew - quantity 14
2. Main bearing cap - quantity 6
3. Lower main bearing - quantity 7
4. Main bearing cap - quantity 1
5. Upper main bearing - quantity 7
6. Thrust bearing - quantity 1
7. Thrust bearing - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Crankshaft bearing rollout tool, Part Number 3823818
- Engine barring tool, Part Number 4919092
- Abrasive pad, Part Number 3823258, or equivalent
- Assembly lubricant, Part Number 3163087
- Dial indicator, Part Number 3376050

- Indicator mounting base, Part Number 3377399

Additional Service Items

- Awl
- Blunt tool
- Outside micrometer that has a ball tip
- Plastic or rubber mallet.

General Information

Two upper halves of the thrust bearings are used in the number 6 main bearing saddle.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)

- Check crankshaft end clearance. Refer to Procedure 001-016 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-016.html)

Remove

⚠ CAUTION ⚠

Do not pry on the main bearing caps to free them from the cylinder block. Damage to the main bearing caps and cylinder block can result.



Note : When replacing main bearing caps in the chassis, replace number 2 through 6 while the number 1 and number 7 caps support the crankshaft. After replacing number 2 through number 6, replace number 1 and number 7.

Loosen the main bearing capscrews completely, but do **not** remove.

Use the main bearing capscrews to "wiggle" the main bearing cap loose. Be careful **not** to damage the capscrew threads.

Remove the main bearing cap.

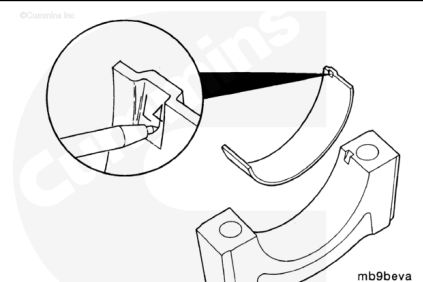
Remove the upper two halves of the thrust bearings. Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/1016/1016-001-007.html)

Mark the main bearings for position and number as they are removed.

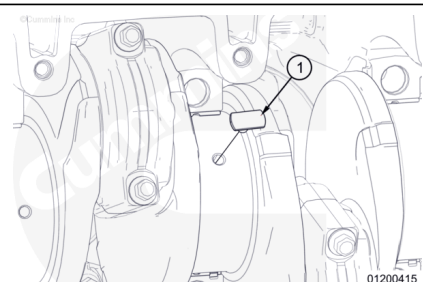
Use an awl to mark main bearing position in the tang area.

Note : Marking main and thrust bearing position is for future identification or possible malfunction analysis.



To remove the upper main bearing, install the crankshaft bearing rollout tool (1), Part Number 3823818, in the oil hole of the crankshaft main bearing journal.

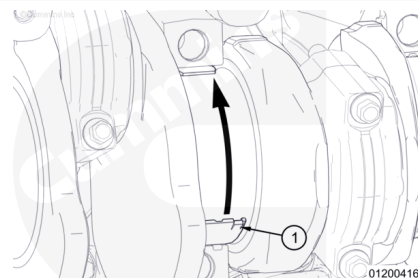
Note : Use the crankshaft bearing rollout tool for all upper main bearings except the number 7 main bearing.



Use engine barring tool, Part Number 4919092, to rotate the crankshaft so the replacer contacts the upper main bearing on the side opposite the tang.

Continue to rotate the crankshaft in the direction that will remove the tang side (1) of the upper main bearing first.

Remove the main bearing.

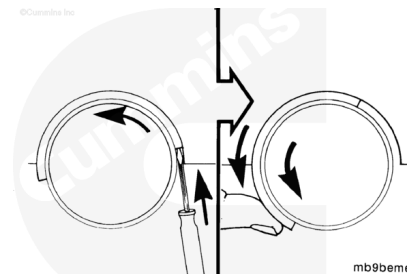


⚠ CAUTION ⚠

Use care not to damage the crankshaft or cylinder block.

Note : The rear main bearing, number 7, does not have a hole in the journal, so the tool can not be used to replace the bearing.

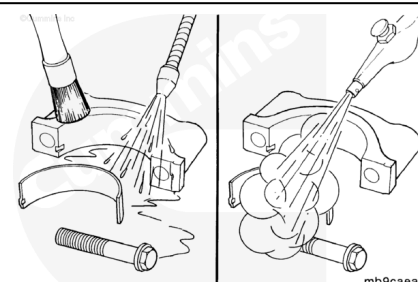
Use a blunt tool to gently bump the end of the bearing to loosen it from the cylinder block. Use finger pressure against the main bearing. Rotate the crankshaft to roll the main bearing out.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

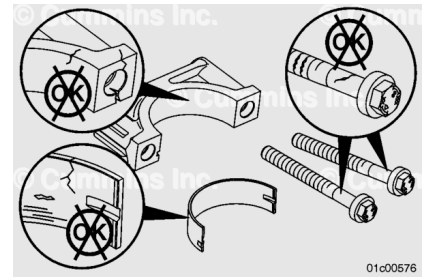
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean or use hot, soapy water to clean the main bearing caps.

Dry with compressed air.

Inspect all main bearing caps, capscrews, and upper and lower bearings for deep scoring, cracks, and so forth.

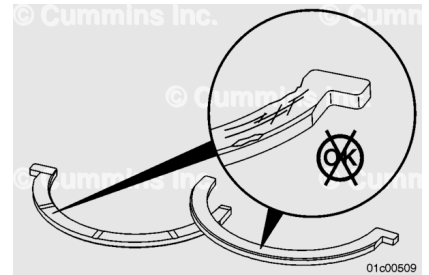
Replace any damaged components. If the main bearing cap is damaged, the block **must** be replaced.



Inspect the thrust bearing for scratches, galling, uneven wear, or other damage.

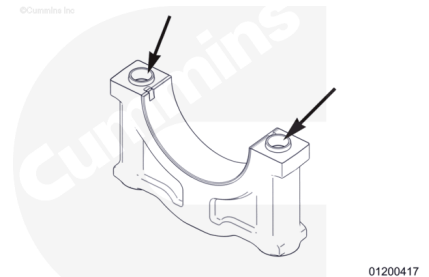
If damage is found on the thrust bearing, replace the thrust bearing.

Inspect the crankshaft, cylinder block, and main bearing cap for damage.



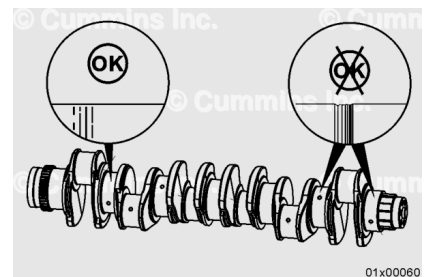
- Refer to Procedure 001-016 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-016.html)
- Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-026.html)

Check the main bearing caps to make sure the ring dowels are installed.



Check the crankshaft main bearing journals for damage or excessive wear. Minor scratches are acceptable.

If crankshaft end clearance measured during Preparatory Steps is found to be out of specification, make sure to check the crankshaft thrust surface for excessive wear or damage. Refer to Procedure 001-016 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-016.html)
Minor scratches are acceptable.

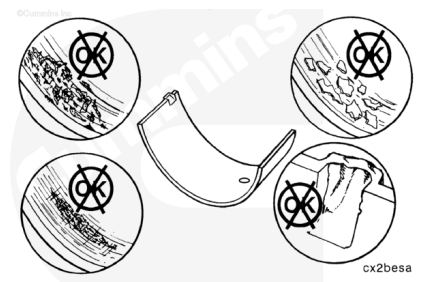


Inspect the main bearings for damage.

Replace any main bearings with the following damage:

- Pitting

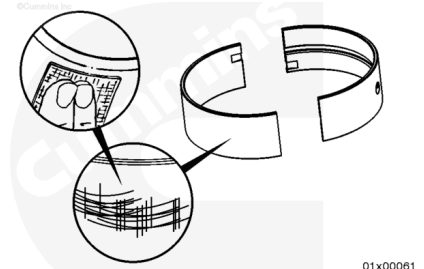
- Flaking
- Corrosion
- Lock tang damage
- Scratches.



Inspect the bearing seating surfaces for nicks or burrs. If nicks or burrs can **not** be removed with abrasive pad, Part Number 3823258, or equivalent, the main bearings **must** be replaced.

Note : If the main bearings are damaged, they **must** be replaced as a set.

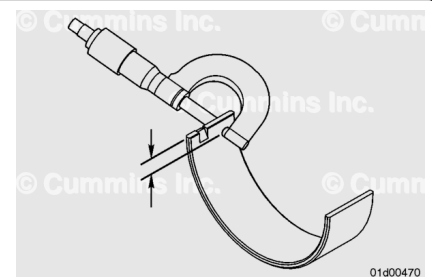
Note : Refer to Service Bulletin, Analysis and Prevention of Bearing Failures, Bulletin 3810387 (/qs3/pubsys2/xml/en/bulletin/3810387.html), for more detailed information on bearing damage.



Measure

Measure the main bearing thickness with an outside micrometer that has a ball tip. This measurement shall be taken at least 5 mm [0.2 in] from the face of the bearing, **not** at the very edge.

Main Bearing Dimensions Standard		
mm		in
3.200	MIN	0.126
3.216	MAX	0.127



If the bearing is **not** within specifications, it **must** be replaced.

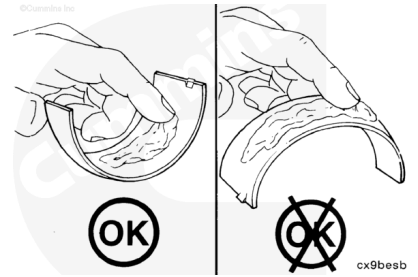
Note : Main bearings are identified with a part number stamped on the back side. They are also identified as lower or upper.



Install

⚠ CAUTION ⚠

Do not lubricate the side against the cylinder block.



Upper Main Bearing

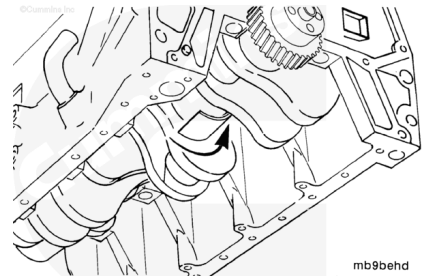
Note : Install the upper main bearing cap after each upper main bearing is installed to keep the main bearing in place while the other upper main bearings are installed.

Apply assembly lubricant, Part Number 3163087, to the upper main bearings.

Note : The upper and lower main bearing shells are not interchangeable. The backs of the main bearings are marked with the proper orientation.

Install the upper main bearings.

Insert the side of the main bearing opposite the tang first in between the crankshaft journal and block. Install the bearing as far as possible by hand.

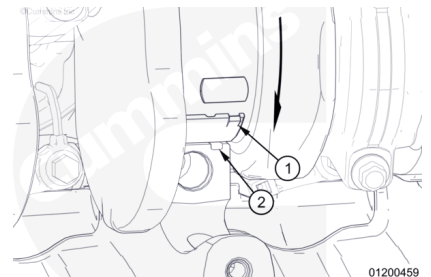


⚠ CAUTION ⚠

Make sure the pin does not slide under the bearing.

Use crankshaft bearing rollout tool, Part Number 3823818, to finish installing the main bearing. Rotate the crankshaft with engine barring tool, Part Number 4919092.

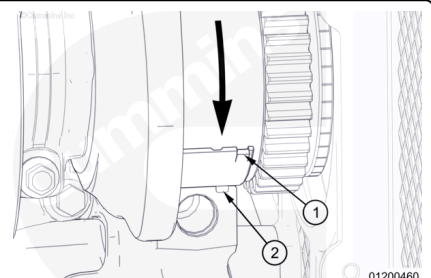
Make sure the tang (1) on the main bearing is located in the notch (2) of the cylinder block. Finish pushing the main bearing into position.



⚠ CAUTION ⚠

Use care not to damage the crankshaft or cylinder block.

Note : The rear main bearing, number 7, does not have a hole in the journal so the pin can not be used to



replace the bearing.

Install the number 7 main bearing.

Insert the side of the main bearing opposite the tang first.

Install as far as possible by hand.

Use a blunt tool to gently push the main bearing into position while rotating the crankshaft.

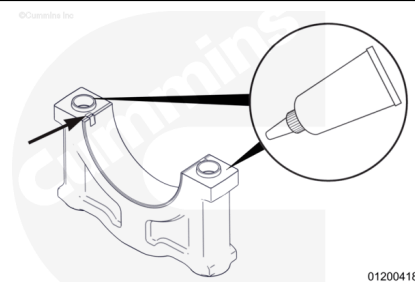
Make sure the tang on the main bearing (1) is located in the notch of the cylinder block (2).

Install the upper two halves of the thrust bearings. Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/1016/1016-001-007.html)

⚠ CAUTION ⚠

Do not lubricate the back side of the bearing that contacts the main bearing cap.



Lower Main Bearings

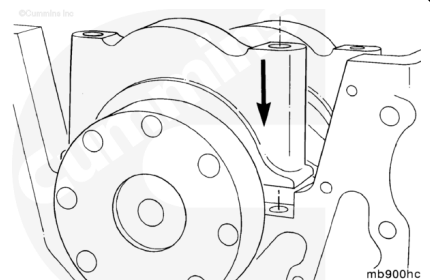
Make sure the backside of the bearings are clean and free of debris. Install the lower main bearings into the main bearing caps.

Make sure to align the tangs of the main bearings with the tangs on the main bearing caps.

Apply a coat of assembly lubricant, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

⚠ CAUTION ⚠

Make sure the main bearing caps are correctly installed in the same position as removed.



Make sure the surfaces between the main bearing cap and block are clean and free of debris.

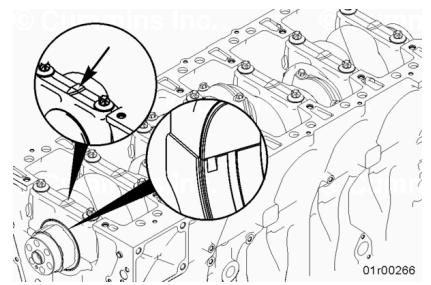
Install the main bearing cap into position, aligning the main bearing cap dowel rings with the cylinder block.

The main bearing caps are numbered before or during the removal process for their location. Number 1 starts with the front of the block.

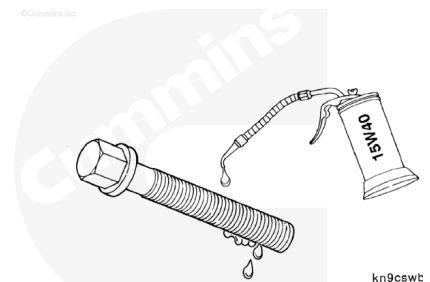
Note : The main bearing caps **must** be installed so the numbers on the caps match the main bearing saddle in the block. The lock tangs in the main bearing saddle and main bearing cap **must** be on the same side.

Install the main bearing caps. Make sure to align the ring dowels on the main bearing cap with the corresponding drillings in the cylinder block.

Arrows on bottom of all caps shall point to front of engine.



Lubricate the main bearing capscrew threads and underside of the head. Use clean engine oil.



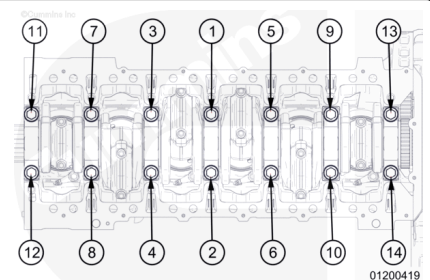
Gently tap the main bearing cap into position with a plastic or rubber mallet.

When seated, install the main bearing capscrews. Tighten.

Use barring tool, Part Number 4919092. The crankshaft **must** rotate freely after installing the main bearing caps.

While applying final torque to the main bearing capscrews, frequently check that the crankshaft rotates freely.

Tighten the capscrews in the sequence shown.

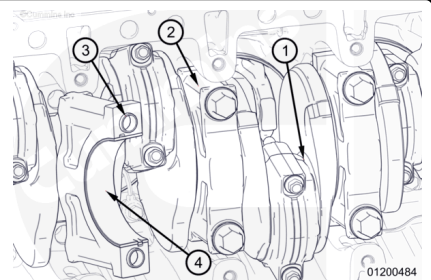


Torque Value:

1. 135 n•m [100 ft-lb]
2. Loosen 720 degrees.
3. 135 n•m [100 ft-lb]
4. 270 n•m [199 ft-lb]
5. Angle 120 degrees.

If the crankshaft does **not** rotate freely:

1. Check if the crankshaft is contacting one of the connecting rods.
2. Check if the correct main bearing caps are installed correctly.
3. Check if the main bearing cap ring dowels or mounting surfaces are damaged during installation.
4. Check if the correct main bearings are installed.



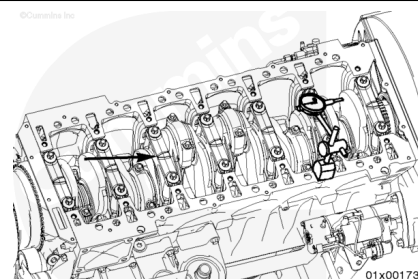
Measure

Use the following steps for the crankshaft end clearance.

To measure the end clearance of the crankshaft, install dial indicator, Part Number 3376050, along with indicator mounting base, Part Number 3377399, onto the oil pan flange.

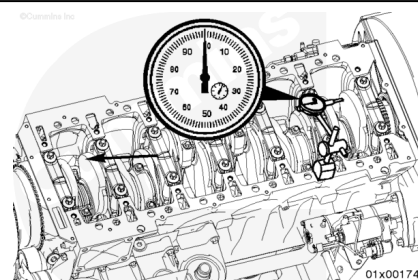
Place the tip of the dial indicator gauge against the crankshaft counterweight.

Push the crankshaft toward the rear of the cylinder block.



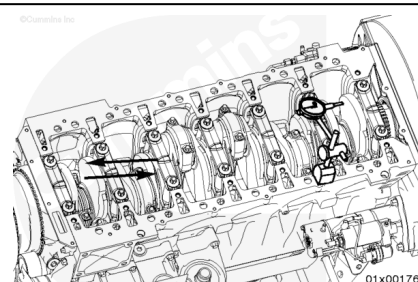
Set the dial indicator at "0" (zero).

Push the crankshaft toward the front of the cylinder block.



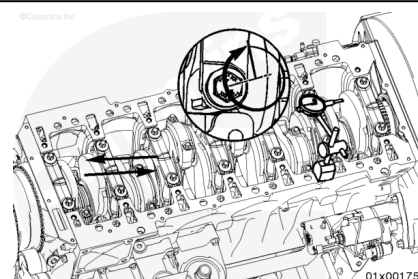
Measure the crankshaft end clearance.

Crankshaft End Clearance - New or Reground		
mm		in
0.16	MIN	0.006
0.36	MAX	0.014



If the end clearance is less than 0.16 mm [0.006 in], do the following:

1. Loosen all main bearing capscrews one turn.
2. Push the crankshaft toward the front and then toward the rear of the cylinder block.

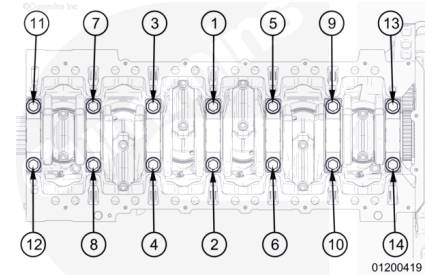


Tighten the main bearing capscrews in sequence.

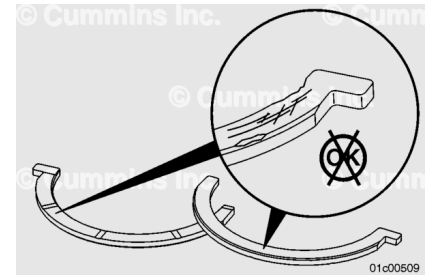
Torque Value:

1. 135 n•m [100 ft-lb]
2. Loosen 720 degrees.

3. 135 n•m [100 ft-lb]
4. 270 n•m [199 ft-lb]
5. Angle 120 degrees.



If the crankshaft end clearance is excessive, remove the thrust bearings and inspect the thrust bearing. Worn bearings can cause excess end clearance. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-007.html>)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html>)
- Install lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.